Modular Language-Guided Robotic Manipulation: Integrating NL-DSL Grounding and Closed-Loop Visual Feedback

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***Abstract*—Traditional imitation learning (e.g., Robomimic [3]) relies on open-loop visual-to-action mappings, lacking structural semantic understanding and robustness against perturbations. We propose extending the Robomimic pick-up task with two innovations: (1) an NL+DSL (Natural Language to Domain-Specific Language) paradigm for precise command parsing, and (2) closed-loop visual feedback for dynamic trajectory regulation. Our system uses a vision-language-action (VLA) model to ground DSL constraints into target objects and 3D grasp poses, which are fed to a low-level policy. By incorporating real-time visual feedback, the policy adjusts actions dynamically. Evaluated on out-of-distribution scenes, we expect our framework to significantly outperform open-loop, pure-NL baselines in generalization and error recovery.**

I. Introduction and Motivation

Recent vision-language-action systems have shown strong potential for robotic control, but robustness in out-of-distribution scenes remains a key challenge [2]. At the same time, task-level progress awareness and recovery are increasingly important for long-horizon manipulation [1].

This project is proposed as a **research project** that builds upon the "General robot pick-up skills learning" idea (2.3) and incorporates elements of "Language-guided motion generation" (2.5) from the idea pool. To improve upon pure open-loop execution and ambiguous natural language parsing, we introduce two key innovations: (1) adopting a Natural Language + symbolic abstraction paradigm inspired by recent LLM-guided planning work [4], and (2) incorporating a closed-loop feedback regulation mechanism during execution, motivated by progress-aware VLA control [1]. We aim to answer the following research question: *Can a modular integration of a pre-trained VLA, utilizing symbolic language grounding and robust visual feedback, improve the generalization of robotic pick-up tasks to novel visual and semantic variations?* By addressing this question, we seek to demonstrate a practical and computationally feasible method for enhancing existing manipulation frameworks with formal grounding and error recovery capabilities.

II. Literature Review

a) VLA Robustness and Test-Time Scaling: Recent work shows that VLA robustness can be improved significantly at inference time by sampling and verification instead of single-pass action generation [2]. This suggests a practical path for course-scale systems: keep the base policy lightweight and improve reliability through test-time decision mechanisms.

b) Modular Planning and Progress-Aware Control: Recent LLM-guided planning work demonstrates that compact symbolic abstractions can improve task planning efficiency and reliability compared to verbose formulations [4]. In parallel, progress-aware VLA control with explicit sub-goals and rewind-style recovery improves robustness in long-horizon manipulation [1]. Beyond manipulation, large-scale motion control studies also support the value of modular, representation-driven control pipelines [5].

III. Technical Plan

A. System Overview

The proposed system consists of four primary modules (Fig. 1):

1) **NL+Symbolic Parsing Module [4]:** A lightweight LLM translates natural language queries into a compact symbolic task representation with clear action constraints.
2) **Semantic Perception Module:** A vision-language grounder uses the DSL and RGB observation to localize targets (e.g., bounding boxes).
3) **Grasp Planning Module:** Computes a 6-DoF target grasp pose to serve as an execution sub-goal.
4) **Closed-Loop Execution & Feedback Module [1]:** A Robomimic policy translates the sub-goal to joint actions, while continuous visual feedback and progress monitoring correct runtime deviations.

B. Implementation Details

a) Baseline Reproduction: We will first reproduce the "Lift" task from Robomimic repositories. Using human demonstration datasets, we will train a BC-RNN policy achieving robust ($\geq 80\%$) success in default scenes to serve as the baseline.

b) VLA Integration via NL+DSL: We will prompt an LLM to translate text queries ("*pick up the red cube*") into rigid JSON-based DSL maps. A lightweight grounder (`Florence-2`) leverages these parsed DSL constraints to extract the 2D spatial region.

c) Planning & Feedback Loop: Given spatial bounds, Robomimic's internal API acts as an idealized 3D grasp planner. Instead of open-loop rollouts, we continuously poll simulator vision states along the horizon. This active check registers relative drift and applies closed-loop policy action corrections with progress-aware triggers [1].

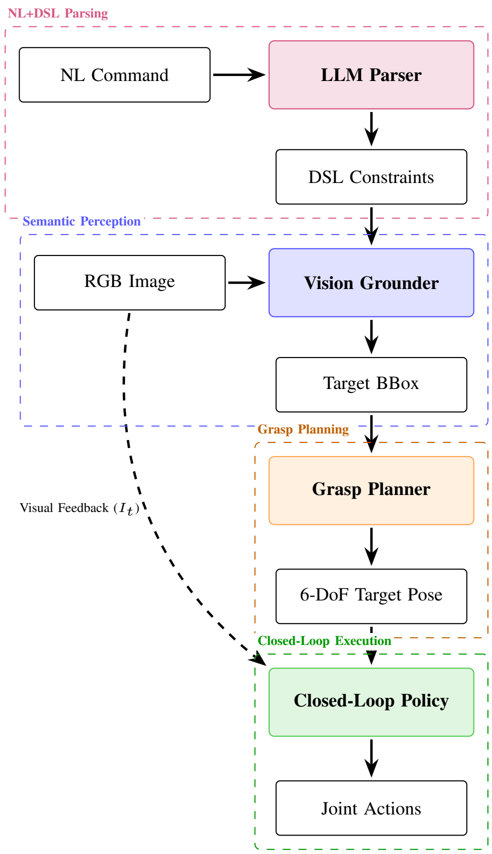

Fig. 1. Proposed modular architecture. An LLM parses the NL command into a DSL. The vision grounder localizes it in the RGB image. The planner generates a 6-DoF pose, executed via a closed-loop policy incorporating visual feedback.

d) Generalization Evaluation: We create distinct test environments containing novel object appearances, patterns, and visual distractors. Success rates across 50 trials per scene will contrast the proposed setup against pure-NL, open-loop baselines.

C. Mathematical Formulation

Let $I_t \in \mathbb{R}^{H\times W\times 3}$ be the RGB observation at time t, and let L be a natural language instruction. The LLM parsing module f_{LLM} converts L into a structured DSL D:

$$D = f_{\text{LLM}}(L). \tag{1}$$

The vision grounder f_{VG} produces a grounding bounding box $\mathbf{b} = (x, y, w, h)$:

$$\mathbf{b} = f_{\text{VG}}(I_0, D). \tag{2}$$

The grasp planner g computes a target grasp pose $\mathbf{p} \in SE(3)$:

$$\mathbf{p} = g(I_0, \mathbf{b}). \tag{3}$$

Finally, the closed-loop Robomimic policy π_θ dynamically generates actions $\mathbf{a}_t$ at each step:

$$\mathbf{a}_t = \pi_\theta(\mathbf{o}_t, I_t, \mathbf{p}), \tag{4}$$

where $\mathbf{o}_t$ is the proprioceptive state, I_t acts as the real-time visual feedback, and $\mathbf{p}$ serves as the execution sub-goal.

IV. Expected Outcomes

1) **Quantitative:** The baseline open-loop policy is expected to achieve a high success rate in the standard environment but drop lower when object colors or backgrounds change. The proposed NL-DSL and feedback-regulated system should maintain a higher success rate across generalization environments.
2) **Qualitative:** A functional demo showing the robot successfully planning and executing grasps specified by complex natural language in dynamic scenes.
3) **Analysis:** A failure mode analysis categorizing errors into LLM parsing failures, grounding inaccuracies, and feedback tracking delays.

V. Potential Risks and Backup Plan

a) Risk 1: LLM latency and Parsing Errors.: Relying on an LLM for NL-to-DSL conversion introduces API latency and potential hallucinated constraints. **Backup:** We can fall back to template-based regex mapping for simple commands or use a quantized local code-LLM to guarantee syntax and reduce round-trip latency.

b) Risk 2: Closed-Loop execution divergence.: Adding active visual feedback I_t to control policies might cause action jitter or cumulative drift. **Backup:** We will apply a low-pass filter over visual feedback corrections or only query visual updates at a lower frequency to ensure robust trajectory following.

c) Risk 3: Simulator ground-truth positions used inadvertently.: To ensure the system remains vision-based, we must avoid querying privileged simulator object positions. **Backup:** If the open-loop grasp sampler fails too often, we will query the simulator's explicit state only for objects bounded by our visual grounder, documenting this clearly in the final report.

VI. Timeline

- **Week 1-2:** Set up Robomimic environment. Reproduce baseline pick-up policy and validate its performance on the standard task.
- **Week 3:** Integrate the NL-to-DSL LLM parser and the lightweight vision grounder (`Florence-2`) into the perception pipeline.
- **Week 4:** Connect vision outputs to the Robomimic policy and implement the closed-loop visual feedback polling mechanism.
- **Week 5:** Design and implement generalization test environments (visual variations). Run experiments and collect success rate data.
- **Week 6:** Analyze results, debug failure nodes, and record the demonstration video.
- **Week 7:** Finalize project report and prepare final presentation.

References

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